

Linux Hardening Audit Tool - Project Report

Introduction

The Linux Hardening Audit Tool is a Python-based cybersecurity project designed to evaluate the security posture of Linux systems. The tool automates security checks and helps identify potential misconfigurations that may expose a system to security risks.

Abstract

This project performs automated security audits on Linux environments by checking firewall status, SSH configuration, file permissions, and running services. The generated report assists administrators in identifying weaknesses and improving overall system security.

Tools Used

Python 3, Linux Operating System, Subprocess Module, Git, GitHub, Terminal Commands, Security Audit Techniques.

Steps Involved in Building the Project

1. Created project structure and Python script. 2. Implemented firewall status auditing. 3. Added SSH configuration verification. 4. Audited critical file permissions such as `/etc/passwd` and `/etc/shadow`. 5. Collected information about active system services. 6. Generated an audit report automatically. 7. Uploaded source code and documentation to GitHub. 8. Documented screenshots and testing results.

Key Features

• Firewall Status Analysis • SSH Security Checks • File Permission Verification • Running Service Enumeration • Automated Audit Reporting

Conclusion

The Linux Hardening Audit Tool successfully demonstrates practical cybersecurity concepts and Linux system auditing techniques. The project improves visibility into system security settings and provides a foundation for advanced compliance and hardening assessments.